
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have advanced rapidly across a wide range of visual question answering benchmarks, spanning visual reasoning, document understanding, and multimodal dialogue. These gains hold across VLMs built on diverse base models, alignment architectures, and training data. Yet recent work reveals a striking blind spot: these same models lag well behind on traditional image classification benchmarks that probe fine-grained visual knowledge. We evaluate a large set of recent VLMs on fine-grained classification benchmarks and pinpoint the factors that drive the disconnect between fine-grained knowledge and other vision benchmarks. Our ablation experiments establish three clear conclusions: a stronger LLM lifts all benchmark scores uniformly, whereas a stronger vision encoder disproportionately boosts fine-grained classification performance. We further show that the pretraining stage is decisive for fine-grained performance, especially when the language model weights are unfrozen during pretraining. Together, these findings chart a direct path toward VLMs with markedly stronger fine-grained visual understanding and vision-centric capabilities.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which integrate vision encoders with large language models (LLMs), have delivered increasingly sophisticated capabilities in interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). These models now handle complex tasks such as open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite this progress, a decisive question has gone unanswered: How well do VLMs perform on fine-grained visual perception, and what does it take to improve them? This paper answers both.

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires models to focus on subtle distinguishing features while remaining robust to intra-class variations in pose, lighting, and background. Traditional vision encoders like CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) deliver strong performance on fine-grained classification benchmarks, yet the performance of newer VLMs on this same task remains largely unexamined—a gap this work directly closes.

Understanding the fine-grained knowledge capabilities of VLMs is essential: perception is the foundation on which advanced capabilities such as understanding and reasoning are built. Real-world deployments of VLMs routinely involve fine-grained, long-tailed distributions where distinguishing between closely related categories is decisive. If a model cannot correctly identify a mushroom species, for example, it has no basis for determining whether that mushroom is poisonous or safe to eat. Just as importantly, pinpointing the factors that govern fine-grained knowledge acquisition directly informs more effective architectures and training strategies for future VLMs.

In this paper, we first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, directly contrasting their performance on fine-grained classification benchmarks with their capabilities on general VLM tasks. We evaluate 15 VLMs—including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), along-

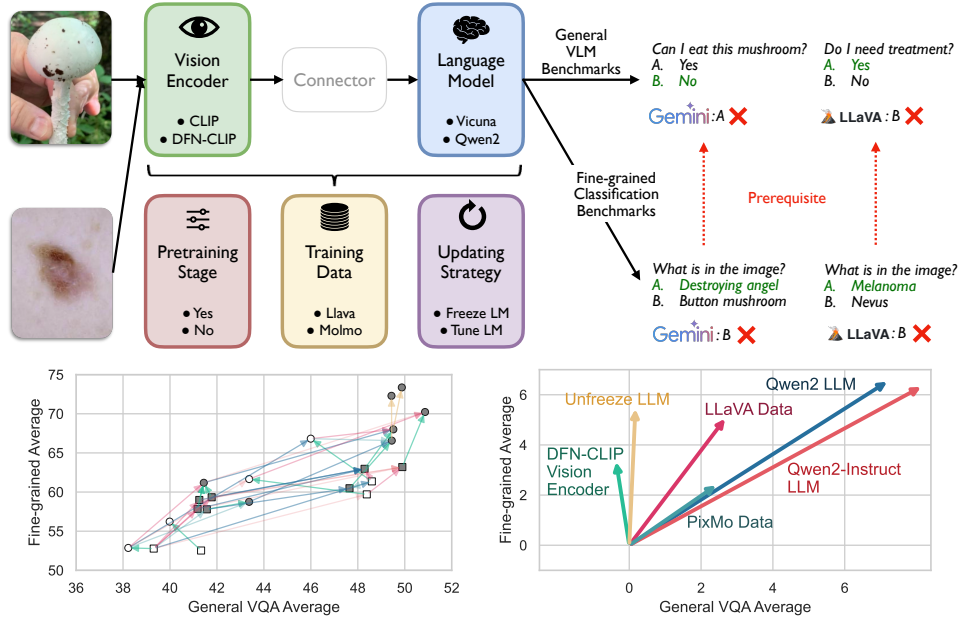


Figure 1: **Overview.** (Top) We investigate the fine-grained classification capabilities of vision-language models (VLMs)—a crucial capability that underpins higher-level understanding and reasoning, yet has been consistently overlooked. (Bottom) Through 22 systematic ablation experiments on key model components and training strategies (left), we isolate the factors that drive fine-grained classification performance in VLMs (right).

side aggregated results from eight general VLM benchmarks such as MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). Our analysis exposes large variance in fine-grained classification performance among models with comparable general abilities, demonstrating that fine-grained recognition is a distinct axis of visual intelligence that existing VLM benchmarks simply do not capture. We further establish a persistent gap between VLMs and their own CLIP-based vision encoders, corroborating and sharpening earlier observations (Zhang et al., 2024; Geigle et al., 2024).

To determine what drives fine-grained classification performance in VLMs, we conduct a systematic study isolating the role of key model components and training paradigms. We disentangle how VLM components—language models and vision encoders—affect classification performance, and how training strategies such as pretraining and data quality shape the outcome. Across 22 carefully controlled ablation experiments, we identify the factors that decisively govern fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Improving the base language model lifts performance uniformly across both fine-grained classification and general VLM benchmarks.
- **Vision encoder (§4.2):** A stronger vision encoder sharply improves fine-grained classification performance while barely moving general VLM benchmarks, particularly in models trained in a two-stage fashion.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning datasets substantially boosts fine-grained classification performance, with only a modest effect on general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification performance rises markedly when both the connector between the vision encoder and LLM and the LLM itself are trained during pretraining, rather than the connector alone.
- **Data quality (§4.3):** The quality of pretraining and instruction-tuning data has little bearing on fine-grained classification, particularly when the language model weights are frozen during pretraining.

In summary, our study puts fine-grained visual perception in VLMs—an essential capability that current benchmark evaluations overlook—squarely on the map. Through systematic ablations, we pin down the key factors that govern fine-grained knowledge acquisition in VLMs. These findings

provide a concrete blueprint for building vision-centric VLMs with stronger visual understanding, directly improving their effectiveness in real-world long-tail tasks that demand fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) are a class of models that integrate visual inputs with language models, typically following the LLaVA-based architecture (Liu et al., 2023) of three core components: a vision encoder, a language model, and an MLP connector bridging the two. These models have achieved strong performance across various multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). To evaluate VLMs, numerous benchmarks have been developed, such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). However, these benchmarks primarily assess reasoning and language understanding given visual inputs, and in doing so they overlook core vision-centric capabilities such as object recognition and fine-grained classification. We close this gap by systematically evaluating and analyzing VLMs on fine-grained classification, an essential aspect of visual intelligence that prior evaluations have left unexamined.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that aims to distinguish between visually similar subcategories within broader object categories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). These benchmarks have played a key role in training and evaluating vision encoders, like ResNet (He et al., 2016), CLIP (Radford et al., 2021) and DINO (Caron et al., 2021). Despite their importance in vision research, recent VLMs have abandoned these datasets in their evaluation protocols in favor of visual question-answering. Because fine-grained classification is critical for real-world applications that involve long-tail distributions and demand precise recognition (e.g., medical diagnosis, food safety, and species identification), evaluating VLMs on such benchmarks is indispensable for gauging their practical usability from a vision-centric perspective, rather than a purely language-centric one.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies that examine VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024) remain narrow in scope, typically covering only a handful of models or datasets. We substantially broaden this line of work, evaluating a far wider range of VLMs across multiple fine-grained classification benchmarks in a unified multiple-choice format. Crucially, we move well beyond prior work with a comprehensive ablation analysis that identifies the key factors driving fine-grained classification performance.

Several recent works have investigated the design space of VLMs, analyzing aspects such as model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). However, these studies concentrate on general benchmarks that are predominantly language- and reasoning-focused. We provide a systematic dissection of how key design choices—base model selection, pretraining methods, and the composition of pretraining and instruction-tuning data—affect fine-grained classification, yielding actionable insights for optimizing VLMs on vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

In this section, we lay out the motivation for evaluating fine-grained classification, describe our benchmarks and evaluation setup, and present the key findings from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a fundamental task in computer vision that focuses on distinguishing visually similar subcategories within a broader category (e.g., different bird species or car models) (Wei et al., 2021). Many evaluation benchmarks, such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have been developed and widely used as standard testbeds

for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) reached near- or superhuman performance on challenging fine-grained classification tasks.

Yet the evaluation of vision-language models (VLMs) has left these benchmarks behind. Although recent VLMs like LLaVA (Liu et al., 2023) interpret visual concepts impressively, their evaluation is confined almost entirely to visual question answering (VQA)—academic VQA benchmarks like MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks like DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks stress reasoning and language processing but contain no fine-grained object categories, leaving core vision-centric capabilities untested.

Fine-grained classification is indispensable for VLMs, because real-world applications routinely encounter long-tailed distributions that demand precise visual discrimination between similar categories. Consider a blind user asking whether a mushroom is safe to eat: the model must correctly identify the species before it can reason about toxicity, and misclassifying a deadly destroying angel as an edible white button mushroom could be fatal. In medical diagnosis, failing to distinguish between visually similar diseases leads directly to incorrect treatment and serious health risks. In self-driving cars, confusing a “stop” sign with a “do not enter” sign—both red and similarly shaped—produces dangerous navigation errors. VLMs must therefore excel at fine-grained classification; nothing less guarantees safety and reliability in real-world deployment.

3.2 BENCHMARKS

We evaluate the fine-grained visual knowledge of VLMs using four well-established object recognition benchmarks, each focusing on different domains of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks span 37 to 1,000 distinct classes, which makes direct evaluation of VLMs in a multiple-choice format—the format used throughout VLM training and evaluation—infeasible in its raw form. We resolve this by adopting the methodology of Geigle et al. (2024) to convert each dataset into 5-way multiple-choice questions. For every image, we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to select the four hardest negative choices by image-label cosine similarity and pair them with the ground-truth label to form the answer set. We deliberately choose the OpenCLIP model to avoid biasing difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023) (all trained on datasets different from OpenCLIP’s). This conversion lets us test VLMs in their native format while preserving most of the original difficulty. CLIP ViT-L/14-336px (Radford et al., 2021), for instance, scores 75.2% on the 102-class Flowers-102 and only 78.3% on our 5-way multiple-choice version—confirming that the task remains hard.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use the respective default prompt format from VLMEvalKit (Duan et al., 2024) and measure accuracy based on the exact match with the multiple choice options. For baseline comparison, we evaluate various CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, where both images and class labels (“A photo of a [class]”) are encoded using CLIP’s image and text encoders. The top-1 prediction is determined by computing the cosine similarity between image and text embeddings.

To determine whether existing VLM evaluation benchmarks capture fine-grained classification capabilities, we compare each VLM’s fine-grained classification performance against its average score on eight general VQA benchmarks, such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), from VLMEvalKit.

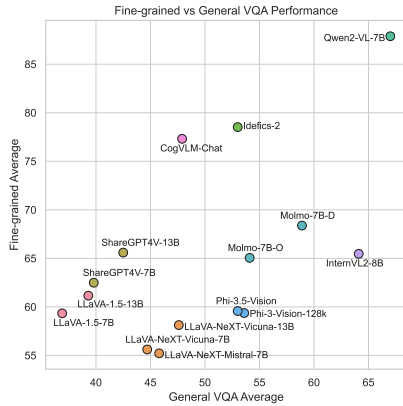


Figure 2: **Fine-grained classification compared to general VQA performance across VLMs.** Our analysis of recent VLMs shows that fine-grained classification is a distinct aspect of visual capability that standard VQA benchmarks fail to measure.

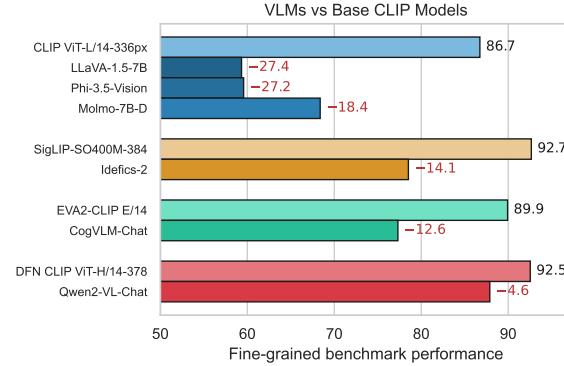


Figure 3: **Comparison of VLMs with their corresponding CLIP vision encoders in fine-grained classification.** Only Qwen2-VL-Chat nearly matches its vision encoder DFN-CLIP; every other VLM falls well behind. This exposes substantial untapped headroom for VLMs on fine-grained classification tasks.

3.4 RESULTS

Figure 2 plots fine-grained classification performance (averaged across our four benchmarks) against general VQA performance (averaged across eight existing VQA benchmarks) for all tested VLMs. Models with nearly identical general VQA performance diverge sharply in fine-grained classification accuracy. CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a), for example, both reach nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracy differs by a full 19pp (percentage points)—77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. This gap establishes that existing VLM evaluation benchmarks do not capture fine-grained classification capabilities, and that fine-grained classification is a distinct dimension of visual intelligence in its own right.

Finding 1: Fine-grained classification is a distinct aspect of visual capability that current VLM benchmarks fail to measure.

Every current VLM builds on an existing pretrained vision encoder. Our multiple-choice format exploits this fact, letting us directly compare VLMs against their own CLIP models on traditional classification benchmarks. Figure 3 makes the comparison for selected VLMs and their corresponding CLIP baselines, and the gap is stark. Molmo (Deitke et al., 2024), a very recent model that excels across VQA benchmarks, reaches only 68.4% average accuracy in our multiple-choice setting, while its own vision encoder CLIP ViT-L/14 (Radford et al., 2021) attains 86.7%. Even the strongest model we test, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points. This gap makes the case unambiguous: we must analyze the design choices behind VLMs to identify the factors that unlock fine-grained classification. The next section does exactly that through a systematic ablation study.

Finding 2: A large, consistent performance gap separates VLMs from their own CLIP vision encoders on fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results in Section 3, we ablate the key differences between models that drive fine-grained classification performance, targeting base model selection and training methodol-

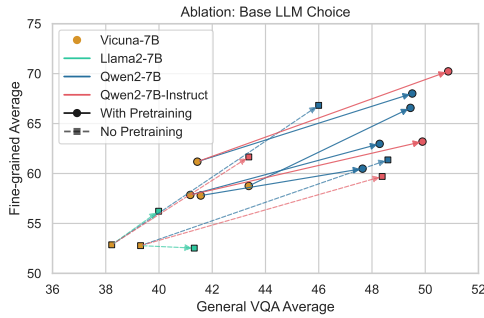


Figure 4: Ablating the base LLM by swapping Vicuna-7B from LLaVA for other LLMs. On average, switching from Vicuna to Qwen2-7B drives a +7.5pp gain in fine-grained performance and a +7.7pp gain in general VQA performance.

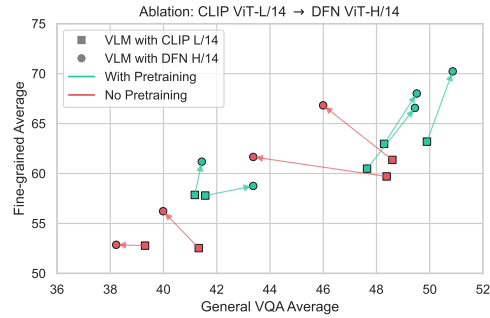


Figure 5: Ablating the base vision encoder by swapping CLIP ViT-L/14 from LLaVA for DFN-CLIP ViT-H/14. Given sufficient pre-training, switching to DFN-CLIP improves fine-grained performance by +4.5pp while also lifting general performance by +1.2pp.

ogy. Our experiments build on the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). Every model undergoes one epoch of pretraining (when applicable) followed by one epoch of finetuning. We hold LLaVA-1.5’s default hyperparameters, including learning rate and batch size, fixed across all experiments, so that any performance difference is attributable to the component we vary.

We measure fine-grained performance as the average score across the four classification benchmarks described in Section 3.2. For general VQA performance, we report the average over three widely-used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). We selected these benchmarks for their broad domain coverage and near-universal adoption in VLM evaluation.

4.2 BASE MODELS

Our analysis in Section 3 shows that recent models typically adopt newer base LLMs, yet most retain the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We now isolate exactly how each of these base model choices affects fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we ablate the LLM component across different vision encoders and pretraining configurations. We pit the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. Figure 4 shows that replacing Vicuna-7B with Qwen2-7B substantially improves performance in every configuration, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct delivers comparable improvements (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base of Vicuna, yields only modest improvements, though the downstream implications of using a model without instruction tuning remain an open question. Critically, all of our trained models—including the lower-performing Vicuna-based VLMs—reliably produce multiple-choice responses in the correct format. The performance differences in fine-grained classification therefore stem from genuinely improved knowledge about the presented options, not from better format adherence.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

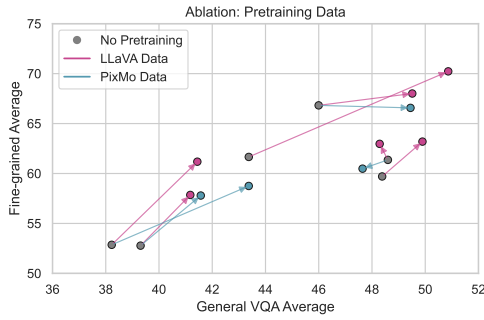


Figure 6: **Ablating pretraining data.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data yields comparable gains. LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp respectively, while PixMo data raises them by +2.4pp and +2.5pp.

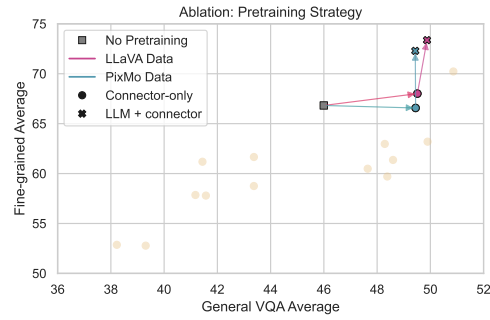


Figure 7: **Ablating the training method: connector-only pretraining versus joint LLM+connector tuning.** Unfreezing the LLM during pretraining delivers a +5.5pp gain in fine-grained performance with no cost to general VQA scores.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which achieves superior zero-shot performance on object recognition tasks like ImageNet. We evaluate both vision encoders across a range of LLM choices and pretraining configurations.

Findings. Figure 5 shows that the effect of substituting CLIP L/14 with DFN-CLIP H/14 hinges on whether connector pretraining precedes finetuning. Without pretraining, DFN-CLIP degrades general VQA performance (-2.5pp) while only modestly improving fine-grained benchmark scores (+2.8pp). Once models undergo connector pretraining, however, general VQA performance holds steady or slightly improves (+1.2pp) while fine-grained performance jumps substantially (+4.5pp). A stronger vision encoder thus delivers a decisive boost to fine-grained classification in properly pretrained VLMs.

Takeaway 2: Better vision encoders (such as DFN-CLIP over CLIP) improve fine-grained classification in VLMs—but only when integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, trains on large-scale image-caption datasets with a captioning objective to align the vision encoder and LLM. The second stage, instruction tuning, refines the VLM on carefully curated multi-round, multi-modal instruction completion data, aligning it with human intent for real-world applications. Here we quantify the impact of each stage on the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We investigate several critical questions about the pretraining process:

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How important is the quality of pretraining data for downstream performance?

We test three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, with

Table 1: **Summary of key ablation settings and comparison to existing models (bolded).** Upgrading the base LLM and vision encoder, and pretraining the connector and LLM on sufficient data, each clearly improve fine-grained performance, whereas instruction finetuning contributes little. General VQA benchmarks show the opposite pattern: there, the choice of LLM dominates.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

high-quality human-annotated detailed captions), and no pretraining. For most experiments, we pretrain only the connector, following Liu et al. (2023). We also examine a strategy similar to Deitke et al. (2024), pretraining the connector for the first 20% of steps before unfreezing the LLM for the remaining 80%. Every configuration is followed by one epoch of finetuning on LLaVA finetuning data.

Findings. Figure 6 presents our pretraining ablation results. Connector-only pretraining consistently improves VLM performance, the sole exception being the Qwen2 and CLIP ViT-L/14 combination. The benefits are especially pronounced for Vicuna-based VLMs, which gain an average of +6.1pp in fine-grained classification and +3.1pp in general VQA performance.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance, with a more modest effect on general VLM benchmarks.

Strikingly, pretraining on low-quality web-scraped captions (LLaVA) matches pretraining on highly detailed human annotations (PixMo): relative to LLaVA data, PixMo data shifts fine-grained performance by just -1.6pp and general VQA scores by +0.4pp.

Takeaway 4: Pretraining data quality has little impact on overall model performance.

We reasoned that connector-only pretraining might prevent the model from fully exploiting higher-quality captions. To test this, we pretrain both the LLM and the connector, using a 20% connector-only warmup phase. Figure 7 confirms that this approach substantially boosts fine-grained classification performance (+5.5pp) without compromising general VQA scores. The improvement holds for both LLaVA and PixMo pretraining data, demonstrating that training on the lower-quality captions does not impair the LLM’s language capabilities.

Takeaway 5: Pretraining both the LLM and the connector substantially improves fine-grained benchmark performance while preserving general VQA scores.

4.3.2 FINETUNING

Setup. Departing from our earlier setup, we begin with the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (from which Qwen2-VL-7B-Chat is trained) and finetune on LLaVA-Instruct data (Liu et al., 2023) following our previous experimental protocol. We compare against Qwen2-VL-7B-Chat, which uses 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison cleanly isolates the effect of the finetuning dataset.

Findings. Finetuning solely on LLaVA finetuning data slightly reduces both fine-grained and general benchmark performance. Fine-grained scores fall by 2.4pp (from 87.9% to 85.5%) and general benchmark scores by 1.3pp (from 62.4% to 61.1%). As Figure 8 shows, this is the smallest contribution to fine-grained performance across all of our ablations, establishing that instruction finetuning

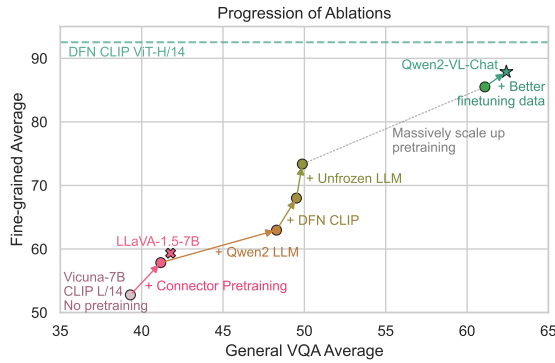


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, we progressively upgrade the base LLM, vision encoder, and training settings, driving fine-grained classification accuracy from 52.8% to 73.4% and closing more than half the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining difference primarily to Qwen2-VL’s extensive 1.4T-token pretraining data.

plays a far smaller role in fine-grained knowledge acquisition than base model selection and pre-training strategy.

Takeaway 6: The instruction finetuning stage contributes far less to fine-grained classification performance than the other factors we study.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablation studies, tracing how each component closes the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Beginning from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021)), we systematically upgrade components to drive fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The picture is clear: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately strengthen fine-grained capabilities relative to general VLM abilities, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for most of the gain in general VQA performance.

These upgrades close most of the gap, leaving a 12-point residual in fine-grained classification unexplained by the ablated components (Table 1). Two factors could account for it: architectural differences and pretraining data scale.

Although architecture could in principle contribute, our observational results (Section 3.4) and training ablations (Section 4.3) point decisively to pretraining data scale as the dominant factor. Our experiments with LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) data used comparatively small datasets—each fewer than 1M images and captions, or roughly 200M and 400M tokens, respectively. Wang et al. (2024b), by contrast, pretrain Qwen2-VL on 1.4T tokens—orders of magnitude more than our experiments. This enormous disparity in pretraining data scale is the most plausible explanation for the remaining gap, underscoring the decisive role of extensive pretraining in building VLMs with superior fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study has practical limitations that define clear directions for future work. Because of computational constraints, we compare training on <1M data points rather than the billion (B) scale training used by some newer VLMs, so the effect of billion-scale training on our observations remains to be measured directly. Newer work also proposes alternative pretraining strategies, which may interact with fine-grained visual understanding in ways we have not yet explored.

486 In this work, we deliver a comprehensive, systematic evaluation of state-of-the-art vision-language
487 models (VLMs) on fine-grained classification benchmarks, establishing fine-grained visual classi-
488 fication as a crucial dimension of VLMs that prior evaluations overlooked. Through an in-depth
489 analysis of key model components and training paradigms, we pinpoint concrete strategies for im-
490 proving fine-grained classification and strengthening vision-centric capabilities—directly expanding
491 the applicability of VLMs to real-world scenarios that demand precise visual understanding.
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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed results of model ablations.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed results of model ablations on general VQA benchmarks.

A FULL RESULTS

We present the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: Detailed results of model ablations on fine-grained classification benchmarks.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As covered in Section 3.2, we evaluate VLMs and base vision encoders on four different traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset covering a broad range of supercategories, each containing multiple fine-grained subcategories, based on the WordNet hierarchy. The ImageNet test set contains 50,000 images with classification across 1,000 classes.

Because of the structure of the ImageNet/WordNet class hierarchy, many classes are too visually distinct from the others to yield hard multiple-choice examples (there are, for instance, only three classes under the “person” category). To address this—and the dataset’s unwieldy size—we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting from 19,056 annotated examples, we narrow the pool to 5,457 images spanning 928 of the original 1,000 classes. Since this curation leaves the remaining classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species characterized by high intra-class variation and inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories with challenging variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories, exhibiting substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For every fine-grained classification test set except ImageNet (described above), we generate hard negatives by running zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top 4 incorrect predictions and shuffle them with the correct label to form the answer choices.

Prompt formatting. In our multiple choice VQA format, the question is always of the form “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n

C. Option 3\n D. Option 4\n E. Option 5”. However, since different VLMs are trained with different multiple choice prompts and format, we use each model’s corresponding format from VLMEvalKit (Duan et al., 2024), for instance: “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, and thus there is no question format. The classnames passed to the text encoder are instead formatted in the template of “a photo of a {classname}”.

B.2 GENERAL VQA

In our initial observational testing from Section 3, we rely directly on the numbers reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). We use this reported average score only in Figure 2.

For all subsequent evaluations—both on existing models and on the models we train in Section 4—we narrow to a focused subset of multiple-choice VLM benchmarks for a cleaner, more direct comparison. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples testing a wide range of VLM capabilities split between perception and reasoning. While the majority of the examples deal with coarse-grained perception and visual reasoning tasks, some examples test “fine-grained perception”, though the categories (such as action recognition, attribute recognition, and OCR) are in a different domain from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a wide distribution of domains and applications. Though the test set has 10,500 examples, most works cite model performance on the validation split, which contains 900 questions. MMMU tests general knowledge tied to visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A dataset comprised of 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, both across domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). Every existing model has a pre-defined evaluation procedure; for our own trained models, we apply the LLaVA evaluation code, since we finetune exclusively on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For the majority of experiments, we adhere to the LLaVA architecture and training procedure.

Pretraining. This means one epoch of pretraining on the LLaVA pretraining data, obtained from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we only use the “caption” field and not the raw audio transcript. During pretraining, we use a batch size of 256. In most of our pretraining runs, we tune only the randomly-initialized MLP connector with a learning rate of 1e-3. When ablating pretraining strategy, we first tune the connector only as warmup, for 20% of steps, then unfreeze the LLM and tune it as well, with a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. For finetuning, we always train for one epoch on the LLaVA-1.5 finetuning data, which consists of LLaVA-Instruct and VQA training data from various datasets to allow proper benchmarking. During finetuning, we maintain a learning rate of 2e-5 for the LLM, with a batch size of 128. The vision encoder remains frozen during both phases of training.

For the ablation experiment on the Qwen2-VL architecture, we start with Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models using 4 nodes of 4 A100 (40GB) GPUs each. Training typically completes within 20 hours, with variation depending on the CLIP model size.

LLaVA vs. PixMo data. In our main text, we refer to the difference in caption quality between LLaVA pretraining data and PixMo pretraining data. LLaVA pretraining data is sourced from CC-3M, which consists of web-scraped captions, commonly known for low quality and relevance. PixMo-Cap is, on the other hand, collected from human annotators recording a 60–90 second audio description of an image, and then combined into a proper image caption (Deitke et al., 2024). A surface analysis of the two datasets makes the contrast concrete: LLaVA captions average 9.8 words, while PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”